

DXFvec BYOK Image Vectorization & DXF Conversion – Full Spec

1. Overview

DXFvec is a web and CLI tool for converting raster images (PNG/JPG/WEBP) into CAD/CNC-ready DXF vectors, with optional AI-powered engines and a bring-your-own-key (BYOK) model for cloud providers. It competes with tools like VectoSolve, Tracepen Online, Vectorizer.AI, DXFai, generic converters (Convertio, CloudConvert, Coolutils, RapidResizer) and desktop software like Scan2CAD.^{[1][2][3][4][5][6][7][8][9][10]}

The default experience relies on local algorithms or open-source AI (no keys required), while advanced cloud AI providers can be plugged in via user-supplied API keys (BYOK).^{[11][12]}

2. Market & Competitive Landscape

2.1 Online converters and AI DXF tools

- **VectoSolve:** AI image→DXF for CNC and lasers; supports PNG/JPG with CUT/ENGRAVE layers and multi-format zips; uses a credit-based pricing model with packs (e.g. 100 conversions) and promises 2–5 second conversion time.^{[9][1]}
- **Tracepen Online:** Browser-based vectorizer that converts PNG/JPG/WEBP to SVG, DXF Lines/Hatch/Faces, PDF; free for basic two-colour vectorization and requires an account for advanced features.^{[13][2]}
- **Vectorizer.AI:** AI-powered image→vector and JPG→DXF conversion with free previews and paid downloads; supports common raster formats and targets CAD/CNC workflows.^{[14][15][3][16]}
- **DXFai:** AI-powered image→DXF and prompt-to-design tool for laser cutting and CNC machining; emphasizes outline generation and engraving designs.^[10]
- **Hyper3D image-to-DXF:** Browser tool to turn logos, scans and sketches into DXF for CAD/CNC fabrication workflows.^[17]
- **Generic converters:** Convertio, CloudConvert, Coolutils and RapidResizer support JPG/PNG→DXF or SVG→DXF conversion but with limited CNC-specific controls and generic UX.^{[4][5][6][7]}
- **ImageToSTL JPG→DXF:** Focused on converting 2D JPG logos/images to DXF model files for AutoCAD and similar CAD applications.^[18]

2.2 Desktop professional software

- **Scan2CAD:** Professional raster-to-vector conversion software; offers batch JPG→DXF conversion, advanced raster cleaning (despeckle, hole filling), object recognition and accurate scaling for technical drawings and maps.^{[19][8]}

2.3 Open-source vectorization engines

- **VTracer (visioncortex/vtracer):** Open-source raster→vector converter that turns images into SVG using efficient algorithms; supports coloured graphics and offers presets (bw/poster/photo) and parameters like corner threshold and speckle filtering.^[11]
- Other GitHub projects under image-vectorization and vectorizer topics provide bitmap→SVG conversion and can inspire AI or advanced algorithms integration.^{[20][21][22]}

3. Product Vision & Positioning

DXFvec is positioned as a free, self-hostable, CNC/laser-centric vectorizer with:

- **Local engines** (Classic and Advanced) requiring no external keys or accounts for core use.^[11]
- **BYOK AI engines** allowing users to integrate commercial AI providers (Vectorizer.AI, DXFai, VectoSolve, etc.) via their own API keys, clearly marked as optional.^{[3][12][1][10]}
- **DXF modes and layers** focused on cutters/engravers: Lines, Hatch, Faces, CUT/ENGRAVE layers and multi-format bundles similar to Tracepen and VectoSolve.^{[2][1]}

The goal is to combine the DXF export richness and UX simplicity of Tracepen, the AI quality of VectoSolve/Vectorizer.AI, and some of Scan2CAD's power, without forcing sign-up or subscriptions for basic use.^{[8][1][2]}

4. User Personas & Use Cases

4.1 Personas

- **Maker / small CNC shop:** Uses LightBurn, RDWorks, LaserGRBL or similar; needs logo and stencil conversions for laser cutting and engraving with reliable CUT/ENGRAVE layers.^{[1][2]}
- **Designer / marketer:** Frequently converts logos, icons and artwork into DXF/SVG for engraving, signage or merch; wants fast, predictable output with minimal CAD knowledge.^{[6][1]}
- **Engineer / architect:** Occasionally converts scanned technical drawings, maps or contour plots to DXF for CAD editing; needs scaling, noise control and topology correctness.^{[19][8]}

4.2 Core use cases

- Convert high-contrast logos, line art and silhouettes from PNG/JPG/WEBP into clean DXF paths suitable for laser cutting and CNC routing.^{[2][3][1]}
- Convert scanned plans, maps and technical drawings into DXF with adjustable noise removal, thresholding and scaling, similar to Scan2CAD workflows.^{[8][19]}

- Export multi-format bundles (DXF, SVG, PNG preview) to support hybrid design + CAD workflows.^{[1][2]}

5. Detailed Product Requirements (PRD)

5.1 Functional scope

5.1.1 Inputs and validation

- Supported raster formats: PNG, JPG/JPEG, WEBP initially (optionally BMP/GIF later), aligning with major online tools.^{[16][14][3][2][^1]}
- File size limit configurable per deployment (default 20–25 MB) with clear error messages and UI hints.^[^1]
- Validation for corrupted or unsupported images with user-friendly feedback.

5.1.2 DXF output modes

- **DXF Lines:** Polylines representing outlines and paths for cutting routers and lasers.^[^2]
- **DXF Hatch:** Hatch entities representing filled/engrave regions for engraving passes.^[^2]
- **DXF Faces:** Closed polylines or 2D faces for 3D modelling tools like SketchUp and Blender.^[^2]
- DXF flavour compatible with AutoCAD 2000 and common CAM software, similar to professional converters.^{[8][1]}

5.1.3 Layer semantics and colouring

- Automatic layer assignment:
 - CUT layer (e.g. colour 1/red) for outer outlines and cut paths.^[^1]
 - ENGRAVE layer (e.g. colour 5/blue) for filled regions or hatch areas.
- Legend in UI explaining layer names and colours, mirroring CNC/laser usage.

5.1.4 Engines and BYOK model

- **Engine types:**
 - **Classic engine:** deterministic pipeline using image processing and contour tracing; always available, no API keys required.
 - **Advanced engine:** local AI-style engine using open-source vectorizers like VTracer; no external keys required.^[^11]
 - **Cloud AI engines (BYOK):** optional providers that require user-supplied API keys, such as Vectorizer.AI, DXFai or similar image→vector APIs.^{[12][3][10][1]}
- **Configuration and UX:**
 - Web UI: dropdown or segmented control Engine: Classic / Advanced (local AI) / Cloud AI (BYOK).

- CLI: flag `--engine classic|advanced|cloud:<provider>`.
- Cloud AI providers are disabled by default and surfaced only when API keys are configured by the operator or user.

5.1.5 Controls and presets

- Basic controls (visible for all engines):
 - Threshold slider.
 - Smoothing slider (curve simplification level).
 - Corner sensitivity (sharp vs smooth corners).
 - Noise filtering level (“remove speckles/holes”), inspired by Scan2CAD’s raster cleaning controls.^[8]
- Advanced controls (visible in an “Expert” section):
 - Minimum region size for vectorization.
 - Maximum path length or node count.
 - Scaling target dimensions (width or height in mm/inches) and output units, matching CAD expectations.^[8]
- Presets:
 - “Logo engraving”.
 - “Laser stencil”.
 - “Technical drawing”.
 - “Contour map”.

Each preset adjusts preprocessing and vectorization parameters to suit typical jobs described for tools like Scan2CAD, VectoSolve and similar converters.^{[1][8]}

5.1.6 Preview and interactive UX

- Side-by-side preview of original raster and vector output with:
 - Zoom and pan.
 - Toggle between raster, pure vector, and overlay modes.
 - Optional display of nodes and paths.
- Legend indicating layer colours and DXF modes (Lines/Hatch/Faces).

5.1.7 Outputs and downloads

- Per-mode downloads: DXF, SVG, PNG preview.
- ZIP bundle download including all selected formats, similar to VectoSolve’s multi-format packs.^[1]
- For Cloud AI engines, surface information about provider (name, cost model, limitations) in a tooltip or info section without leaking keys.

5.2 Non-functional requirements

- **Performance:** Typical logo-size images should convert in <3–5 seconds on standard cloud hosting, comparable to AI-based online converters.^{[9][1]}
- **Reliability:** High success rate for valid inputs; robust error handling and clear feedback on failure.
- **Compatibility:** DXF outputs tested manually in LightBurn, AutoCAD, Fusion 360, VCarve, Inkscape, RDWorks, LaserGRBL, SketchUp and Blender.^{[2][8][^1]}
- **Security and privacy:** Public instances should not persist user images after conversion; BYOK secrets stored securely (environment variables or encrypted config).
- **Accessibility:** Keyboard-friendly controls, high-contrast UI, responsive layout for desktops and tablets.

5.3 Success metrics

- Average conversion time and resource usage by engine.
- Percentage of DXF files that open without warnings in target software (QA metric).
- User feedback scores on “ready-to-cut” quality and ease of use.
- Adoption rate of Advanced and Cloud AI engines vs Classic engine.

6. Technical Requirements & Design (TRD)

6.1 Architecture

DXFvec uses a modular architecture with distinct responsibilities for preprocessing, vectorization, DXF writing, pipeline orchestration, providers, CLI and web frontends.

- **Frontend surfaces:**
 - Web application served by `web.py` (FastAPI/Flask/Streamlit or similar).
 - CLI executable powered by `cli.py` with subcommands for single and batch conversion.
- **Backend pipeline:**
 - Core Python modules `preprocess.py`, `vectorizer.py`, `dxf_writer.py` and `pipeline.py` for deterministic and AI-style vectorization flows.
- **Providers layer (BYOK):**
 - `providers.py` defines interfaces for Classic, Advanced (local AI) and Cloud AI engines, with configuration for keys and options.

6.2 Module designs

- `preprocess.py`:
 - Input validation and format checks.
 - Conversion to grayscale or colour spaces as required.
 - Thresholding (global, adaptive) and noise removal (speckle filtering, hole filling) similar to raster cleaning operations in professional converters.^[8]
- `vectorizer.py`:
 - Contour extraction and path generation using image processing algorithms.
 - Smoothing and corner handling with configurable thresholds.
 - Labeling of paths as outline versus fill regions.
- `dxf_writer.py`:
 - Mapping of vector paths to DXF Lines/Hatch/Faces entities.^[2]
 - Layer assignment for CUT/ENGRAVE semantics, colour codes, line types.^[1]
 - Ensuring AutoCAD-compatible DXF structure and testing across major CAD/CAM tools.^[8]
- `pipeline.py`:
 - High-level API `convert_image_to_dxf(image_bytes, config)` that plugs in a selected provider and returns DXF/SVG/PNG outputs.
- `providers.py` (BYOK engine hub):
 - **ClassicProvider**: wraps deterministic pipeline (preprocess + vectorizer + dxf_writer).
 - **VTracerProvider**: integrates VTracer to convert images to SVG, then passes paths to `dxf_writer` for DXF export.^[11]
 - **CloudAIProvider** implementations:
 - **VectorizerAIProvider**: uses Vectorizer.AI's API via Python SDK to obtain SVG/DXF, controlled by user API key.^{[3][12]}
 - **DXFaiProvider**: calls DXFai's image→DXF API when keys are configured.^[10]
 - Additional providers for VectoSolve or Hyper3D if their APIs are accessible.^{[17][1]}

Each Cloud provider reads keys from environment variables or configuration files and is disabled when keys are absent, aligning with BYOK.^[12]

- `web.py`:
 - HTTP endpoints: `/upload`, `/convert`, `/download/<id>`.
 - Engine selection and presets passed to backend via JSON payload.
- `cli.py`:
 - Commands like `dxfvec convert input.png --engine classic --mode lines --out output.dxf`.

- Batch mode: `dxfvec batch ./input_dir --engine advanced --mode lines,hatch --out ./out_dir.`

6.3 Data flow with BYOK engines

1. Web or CLI receives image and configuration (engine, mode, controls).
2. `pipeline.py` selects provider based on engine and availability (Classic/Advanced/Cloud). If Cloud provider lacks a key, it falls back to Classic or Advanced.
3. Provider runs preprocessing and vectorization (local algorithms or remote AI), yielding vector paths and metadata.
4. `dxf_writer.py` converts paths and metadata into DXF entities with appropriate layers and modes.
5. Output bundle (DXF/SVG/PNG/ZIP) is returned to frontend and offered for download.

6.4 Performance and scaling

- Deterministic engines optimized to avoid unnecessary copies and re-allocations.
- Cloud AI providers integrated with timeouts and circuit breakers; fallback to local engines on failure.
- Horizontal scaling via containerization (Dockerfile and docker-compose) and cloud configs (render.yaml, vercel.json, Procfile) already present in the repo.

6.5 Security and BYOK handling

- API keys managed via environment variables, secrets managers or encrypted configuration files; never logged.
- Web UI should never expose keys; toggles merely indicate provider availability.
- Public demo deployments should use only Classic/Advanced engines to avoid sharing private credentials.

6.6 Testing and QA

- Unit tests for preprocessing, vectorization, DXF writing, and provider selection logic.
- Golden-file tests: given canonical test images, verify the structure and topology of outputs across engines.
- Manual QA in CAD/CAM software to ensure DXF compatibility and layer semantics.

7. AI Integration Design (Local & BYOK)

7.1 Local AI via VTracer

- Integrate VTracer using Python bindings to obtain high-quality SVG paths from raster images.^[^11]
- Use VTracer presets and parameters (corner threshold, speckle filtering) to implement Advanced engine settings and presets.
- Convert SVG paths to DXF entities via `dxf_writer`, preserving colour or layer information for CUT/ENGRAVE semantics.

7.2 Local ML (segmentation-based)

- Optional integration of lightweight segmentation models (e.g. U-Net variants) to produce masks for outlines and engrave regions, followed by contour tracing.
- This approach mirrors AI DXF tools that emphasize outline generation and engrave design extraction.^{[17][10]}

7.3 BYOK Cloud AI providers

- CloudAIProviders interface wraps external APIs like [Vectorizer.AI](#), DXFai or similar image→vector services.^{[3][10]}
- Each provider exposes configuration for endpoint, key, timeouts and output formats.
- Engine selection logic routes to these providers only when a valid key is configured and the user chooses Cloud AI.

8. Documentation & User Guidance

8.1 User documentation

- Explain supported input formats and best practices (high-contrast logos, cleaned scans, adequate resolution), similar to tips found on VectoSolve and [Vectorizer.AI](#).^{[3][1]}
- Document DXF modes and layer semantics for common CNC/laser workflows.^{[2][1]}
- Provide guidance on choosing engines: Classic vs Advanced vs Cloud AI, including trade-offs of speed, quality and privacy.

8.2 Operator documentation

- Explain how to configure BYOK providers, including environment variables and secret management.^[^12]
- Describe resource requirements and scaling strategies for local AI/ML engines.

9. Summary

This report specifies DXFvec as a multi-engine, CNC-centric image→DXF tool with a default local pipeline and optional BYOK AI integration. It draws on patterns from leading tools such as VectoSolve, Tracepen Online, [Vectorizer.AI](#), DXFai, Hyper3D, generic file converters and Scan2CAD for feature and UX design. DXFvec's differentiator is the combination of self-hostability, key-free baseline functionality, explicit BYOK AI support, CNC-aware DXF layering and a unified web/CLI pipeline.^{[13][10][17][3][11][8][1][2]}

References

1. [Image to DXF Converter — CNC & Laser-Ready Files ...](#) - Turn any PNG or JPG into a clean DXF for CNC, laser, or plasma. AI traces precise cut paths with CUT...
2. [Tracepen Online: Free Image Vectorizer — SVG, DXF, PDF & CNC](#) - Tracepen Online converts PNG, JPG, or WebP to SVG, DXF, and PDF right in your browser — free for 2-c...
3. [JPG to DXF Converter | Image to DXF for AutoCAD & CNC - Vectorizer.AI](#) - Convert JPG images into DXF vectors for CAD and CNC workflows. AI tracing, free preview, editable ou...
4. [Convert PNG to DXF Online — Free Image to CAD Converter](#) - Convert PNG images to DXF format for AutoCAD, CNC routers, and laser cutters online. Free, no regist...
5. [Free Online Raster to Vector Converter](#) - Automatically convert a picture to a PDF, SVG, DXF, AI, or EPS vector drawing. Trace outer- or cente...
6. [JPG to DXF — CAD Converter](#) - Convert JPG to DXF for free on [convertio.co](#). Transform raster images into AutoCAD-compatible DXF dra...
7. [SVG to DXF Converter](#) - SVG to DXF Converter - CloudConvert is a free & fast online file conversion service.
8. [Convert JPG to DXF | Professional Conversion Software](#) - Convert JPG to DXF with Scan2CAD. Professional-grade CAD conversion software. Try for free. Download...
9. [Image to DXF Converter Free Online 2026 - VectoSolve](#) - Convert PNG or JPG to DXF for CNC machines and laser cutters in 2-5 seconds. AI creates precise vect...
10. [Image to DXF Converter](#) - Convert any image to DXF vector files for laser cutting and CNC machining. AI-powered outline genera...
11. [visioncortex/vtracer: Raster to Vector Graphics Converter - GitHub](#) - visioncortex VTracer is an open source software to convert raster images (like jpg & png) into vecto...
12. [Python SDK for Vectorizer.AI - GitHub](#) - Python SDK for [Vectorizer.AI](#). Contribute to mitchbregs/vectorizer-ai development by creating an acco...
13. [Tracepen Online — Chuyển Ảnh Sang SVG, DXF Miễn Phí](#) - Tracepen Online – chuyển ảnh raster sang vector sạch ngay trên trình duyệt. Xuất SVG, PNG, PDF, DXF ...
14. [Images से DXF ऑनलाइन कन्वर्ट करें - इमेजिस को CAD, CNC, लेज़र कटिंग और टेक्निकल ड्राइंग के लिए संपादन करने योग्य DXF वेक्टर में बदलें | A...](#)
15. [Konwerter obrazu na DXF | Konwertuj obrazy na DXF dla CAD i CNC](#) - Konwertuj obrazy na edytowalne wektory DXF dla CAD, CNC, cięcia laserowego i rysowania technicznego....

16. [Converti Immagini in DFX per CAD e CNC - Vectorizer.AI](#) - Converti immagini in vettori modificabili DFX per CAD, CNC, taglio laser, e disegno tecnico. Support...
17. [Image to DXF Converter - Convert Photos to DXF](#) - Convert images to DXF with Hyper3D. Turn logos, scans, and sketches into CAD-ready output for CNC, l...
18. [Convert JPG to AutoCAD DXF Format for Free](#) - Convert JPG to DXF AutoCAD with our free and fast online tool. Convert your 2D JPG image or logo int...
19. [Scan2CAD Regular](#)
20. [GitHub - delbeke/vectorizer: Convert bitmap to SVG vector image](#) - Convert bitmap to SVG vector image. Contribute to delbeke/vectorizer development by creating an acco...
21. [Build software better, together](#) - GitHub is where people build software. More than 150 million people use GitHub to discover, fork, an...
22. [vectorizer · GitHub Topics](#) - A java based vectorization software which takes a raster image as input and convert it to a vector f...